Harsh Awasthi

harsh.awasthik@gmail.com • (+91) 9336259023 • harshawasthi.in • linkedin.harshawasthi.in • github.harshawasthi.in

RESEARCH & PROFESSIONAL EXPERIENCE

AI Engineer Intern (Remote) **Nov 2025 – Present**
- SourceworxAI *Noida, UP, India*
- Architected and evaluated core components for an enterprise AI agent platform supporting dynamic tool execution, agent orchestration, and fine-tuning/quantization workflows.
- Engineered full-stack agentic systems tailored to operational automation for non-profit and social enterprise use cases.
- Deployed an on-premise document intelligence pipeline utilizing vLLM, GLM-OCR, and Gemma for automated extraction and structuring of highly sensitive private records.

Technical Intern **Jun 2026 – Jul 2026**
- Andromeida Maritime Solutions *New Delhi, India*
- Designed and implemented a modular mission planner for AUVs, ASVs, and field robotics platforms.
- Constructed a robust communication pipeline using TCP, UDP, WebRTC, and Protocol Buffers featuring custom heartbeat and handshake protocol semantics.
- Integrated hardware-software stacks to execute waypoint-based navigation algorithms, achieving ~95% path accuracy in field test scenarios.

Research Intern **Dec 2025 – Jan 2026**
- Indian Institute of Technology (BHU) *Varanasi, UP, India*
- Accelerated pre-processing of high-resolution remote sensing imagery via PCA-based dimensionality reduction and feature compression.
- Developed an end-to-end 3D reconstruction pipeline using SMAL, SD-Fit, and ViT feature extractions to estimate single-image 3D pose, shape, and camera parameters.

EDUCATION

Bachelor of Technology in Computer Engineering **2023 – 2027**
Aligarh Muslim University *Aligarh, UP, India*
Cumulative GPA: 8.93 / 10.0

SKILLS

Programming Languages & Web Frameworks: Python, C/C++, SQL, JavaScript, FastAPI, Django REST Framework, React, Next.js, Node.js.
AI / ML & LLM Systems: PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, RAG, Agent Orchestration, Model Fine-Tuning, Quantization, OpenCV, Scikit-Learn, CNNs, Vision Transformers, NLP.
Systems & Observability: Docker, Linux, vLLM, OpenTelemetry, Grafana, Prometheus, Loki, Tempo.
Robotics & Networking: TCP/IP, UDP, MQTT, WebRTC, Protocol Buffers, Edge Computing Systems.

PUBLICATIONS & PRESENTATIONS

- **From Rules to Pixels: A Decoupled Framework for Segmenting Human-Centric Rule Violations** 2025

Formulated a neuro-symbolic framework coupling LLM rule interpretation with visual primitives for violation segmentation; curated the HRS benchmark (1,100 images) and achieved **+19.4% mIoU improvement over CLIPSeg**. Published in **NeurIPS MusIML Workshop, 2025**.

- **Hybrid CNN-Transformer Network with Feature-Level Fusion for Automated Classification of Cervical Cytology** 2026

Designed a hybrid CNN–Vision Transformer architecture fusing four backbones with an MLP classifier and Grad-CAM interpretability, reaching **99.34% accuracy and 99.23% F1-score** on SIPaKMeD dataset. Presented at **IEEE DELCON, 2026**.

LEADERSHIP & SERVICE

Computer Team Lead **May 2024 – May 2026**

AUV-ZHCET Club, Aligarh Muslim University *Aligarh, UP, India*

- Directed a team of 10+ student engineers across perception, navigation, and embedded control modules for autonomous marine vehicles.
- Oversees system integration, hardware-in-the-loop testing, and inter-departmental collaboration for international robotics competitions.

PROJECTS

Air Quality Estimation via Satellite Imagery **Jul 2025 – Aug 2025**

- Built a pan-India PM2.5/PM10 estimation framework by fusing satellite, meteorological, land-use, and demographic sources using ensemble strategies (Random Forest, XGBoost, LightGBM, GAM).
- Optimized hyperparameter tuning via Optuna, SHAP, and permutation feature selection, achieving an **80% R^2 accuracy score** across multi-year prediction models.

AGENDA AI: Personal Productivity Agent Engine **Dec 2025**

- Engineered an autonomous productivity assistant with LangGraph and Model Context Protocol (MCP) to manage schedules across Google Workspace and GitHub via natural language.
- Constructed a collaborative Streamlit dashboard integrating task planning, OAuth 2.0 flows, and Langfuse observability for execution tracing and token budget monitoring.

NewsSure: Misinformation Verification Engine **Oct 2025 – Dec 2025**

- Developed a full-stack platform (React + Django REST) incorporating image verification, web scraping, and embedding semantic similarity with stance detection to evaluate automated news credibility.

Autonomous Underwater Vehicle (AUV) Perception & Control **Aug 2024 – Feb 2025**

- Architected software modules for visual localization (SuperPoint, LightGlue, PnP), SLAM, and PID depth control on Raspberry Pi/Jetson hardware. Presented findings at IEEE Delcon 2025.

Autonomous Surface Vehicle (ASV) Navigation Stack **Apr 2026 – May 2026**

- Implemented a PyGame ground station control layer and TCP/UDP communication pipeline. Implemented carrot-following waypoint navigation with tuned PID loops for precise path tracking on Jetson platforms.

HONORS & AWARDS

National Finalist, Bharatiya Antariksh Hackathon (BAH): Selected among top 28 teams nationally out of 1,000+ applicants by ISRO (2025).

SSGSA Scholar: Awarded prestigious competitive mentorship scholarship for academic and research excellence (2026 Batch).

1st Place Winner, Ignite Hackathon: Awarded 1st place by Starlabs for creating an automated AI news verification agent platform (2025).

1st Place (2026) & 2nd Place (2025), AMULATE (BMW Group): Recognized for building autonomous agentic AI assistants and car recommendation workflows.

2nd Place, AMU Underwater ROV Challenge: Engineered PID control systems for a 4-DOF underwater remotely operated vehicle (2024).

Best Innovation Award, MTS Tech Symposium (IIT Madras): Honored for vision-capable sustainable autonomous underwater vehicle design (2025).

Global Finalist, Singapore AUV Challenge (SAUVC): Shortlisted among top 100+ global international university teams (2025, 2026).